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8. Methanol, ethanol and butanol all belong to the homologous series of alcohols.

(a) The table shows information about two different methods for the manufacture of ethanol.

	Method **A** fermentation	Method **B** addition reaction
Raw material	sugar from sugar cane	ethene from crude oil
Reaction	yeast catalyst $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$	phosphoric(V) acid catalyst $C_2H_4(g) + H_2O(g) \rightarrow C_2H_5OH(l)$
Operating pressure	1 atm	60 atm
Type of process	batch (stop-start)	continuous (runs all the time)

Use the information in the table and your knowledge to answer the following question.

Explain **two** advantages and **two** disadvantages of method **A** compared with method **B**. [4]

Advantages

1. ....

....

2. ....

....

Disadvantages

1. ....

....

2. ....

....

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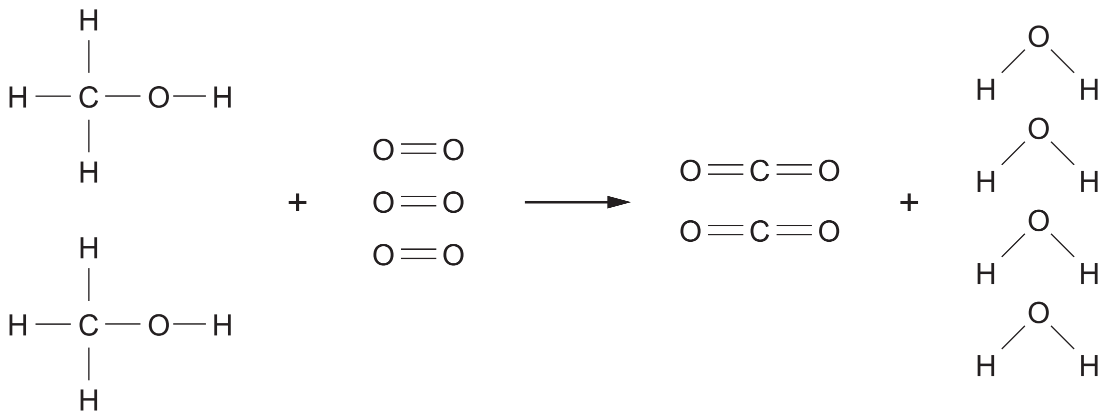
(b) Ethanol is found in alcoholic drinks. Alcoholic drinks turn sour when left exposed to air because ethanol is oxidised to ethanoic acid and water.

Complete the balanced symbol equation for this reaction. [1]

$C_2H_5OH \ + \ O_2 \longrightarrow$ ....

(c) Methanol readily burns in air. The diagram shows the bonds which are broken and the bonds which are formed during the combustion of methanol.



Some relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
C—H	413
C—O	358
C═O	805
O—H	464

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(i) The total energy needed to break all the bonds in the reactants is 5616 kJ. The energy needed to break the bonds in one molecule of methanol is 2061 kJ.

Use this information to calculate the amount of energy needed to break **one** O═O bond. [2]

Energy needed = ... kJ

(ii) Calculate the total energy released when all the bonds in the products are formed. [2]

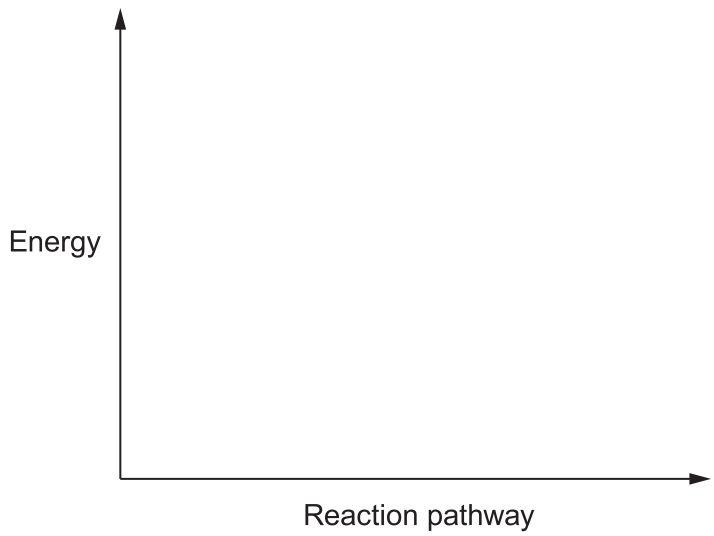
Energy released = ... kJ

(iii) The burning of methanol gives out heat and is said to be exothermic. Use the total energy value 5616 kJ and your answer to part (ii) to show that this is correct. [1]

....

....

(iv) On the axes below draw the energy profile for the combustion of methanol and use the symbol ($\updownarrow$) to show the activation energy for the reaction. [1]



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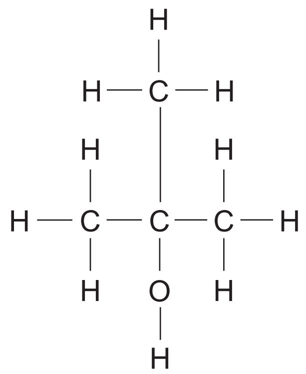
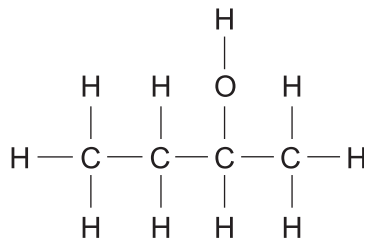
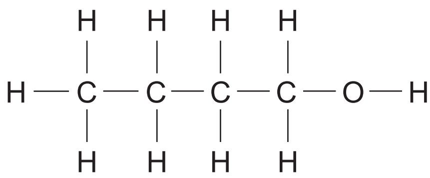
(d) Butanol has the molecular formula C_4H_9OH. It has four positional isomers, **A**, **B**, **C** and **D**.



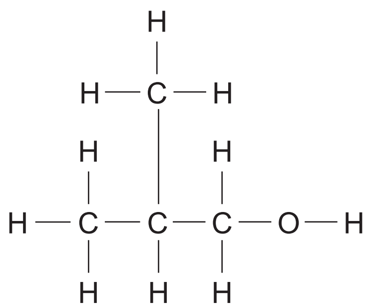
A



B



C



D

Give the **letter** of the isomer corresponding to each of the names in the table. [2]

Name	Isomer
butan-1-ol	
butan-2-ol	
2-methylpropan-1-ol	
2-methylpropan-2-ol	

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